



**DoubleML**

# DoubleML - Recap

DoubleML Workshop

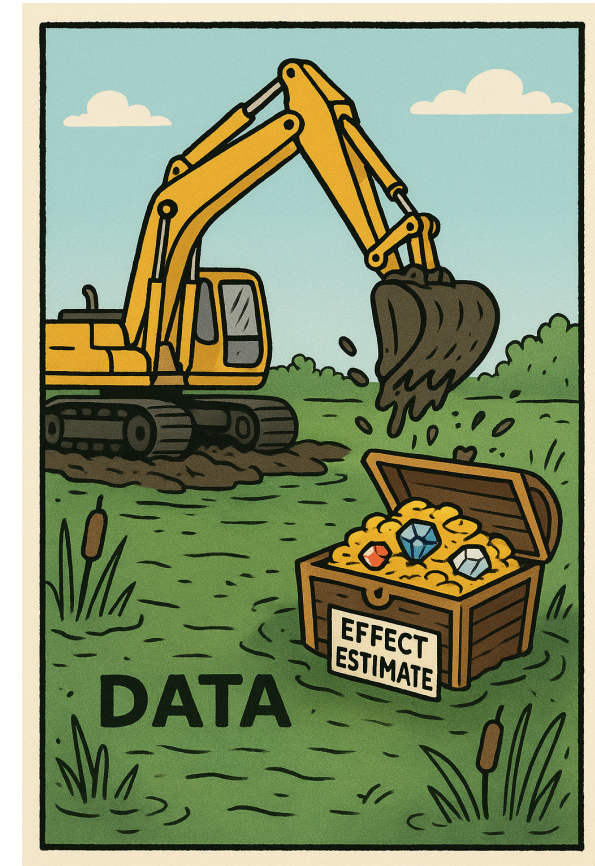
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Economic AI

# Motivation

# Recap / Takeaways from Day 1

- We started from the goal to make causal statements based on data
- Causal inference is concerned with **identification** and **estimation** of causal effects.
- There's a long history of methods for causal inference in statistics, econometrics, epidemiology, and other fields.
  - Current textbooks: Cunningham (2021), Huber (2023)



Looking for causality in a swamp of data.

# Recap / Takeaways from Day 1

- Causal interpretation “*comes*” from **identification**, which is obtained from our assumptions, for example
  - Independence in A/B tests
  - Unconfoundedness in observational studies
  - Exclusion restriction in IV settings
  - Parallel trends in DiD settings
  - ...

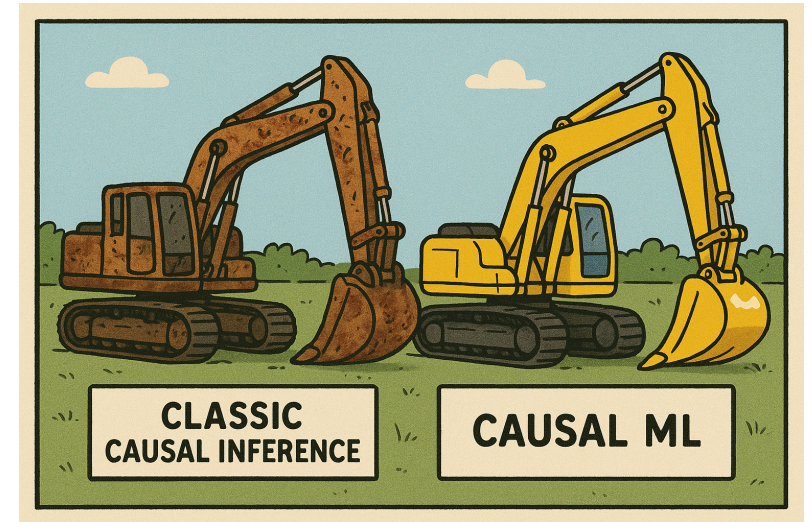


Causality *comes* from identification. Under which assumptions can we interpret effect estimates *causally*?



# Recap / Takeaways from Day 1

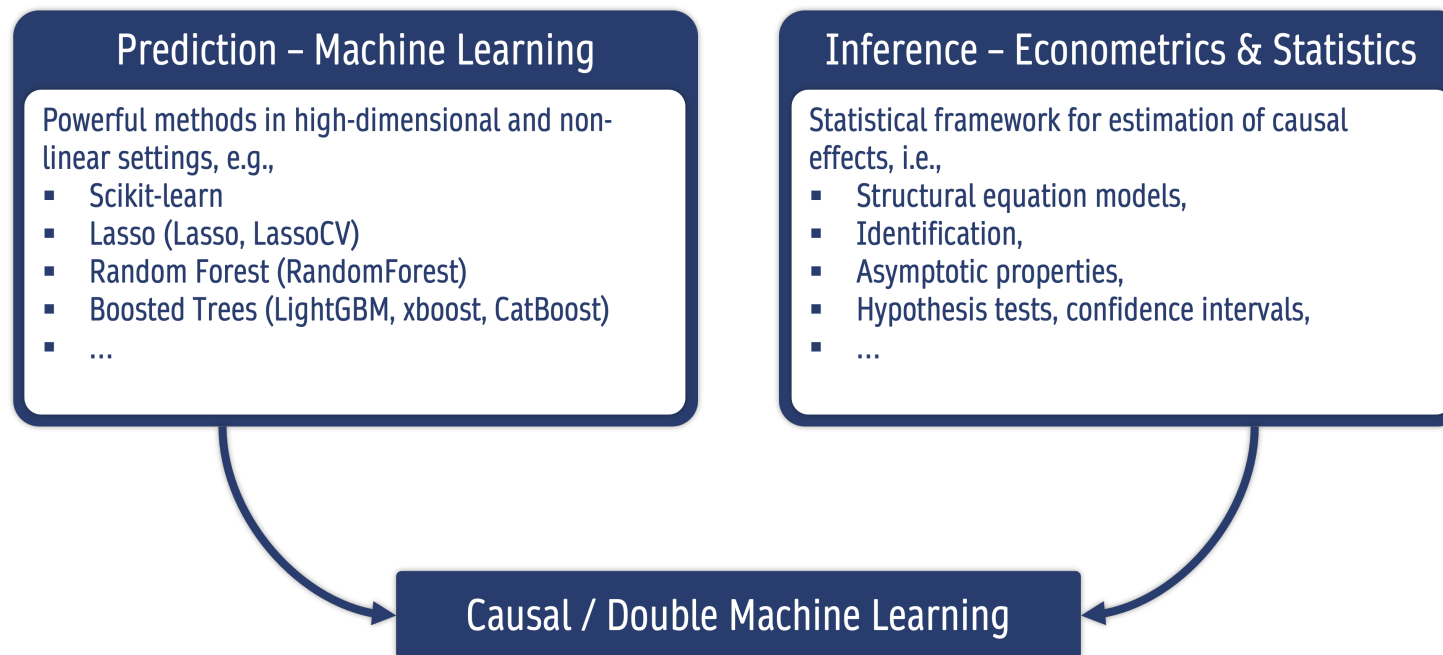
- Causal ML is the **strengths** of ML for causal analysis
  - Flexible modeling without strong parametric assumptions (like linearity)
  - High-dimensional data (many covariates, text/image data)
- To use ML for causal inference, we need to address typical ML challenges
  - Straightforward/naive ML use can lead to systematic biases of the causal estimate (even despite given identification!)
  - We have to handle **regularization** of ML methods
  - **Neyman orthogonality** is a key concept to address regularization bias (motivating example: FWL PLR)



Causal ML: Using powerful ML techniques for causal inference.

# What is Double/Debiased Machine Learning (DML)?

- DML is a general framework for **causal inference** and estimation of causal parameters based on **machine learning**
- Summarized in Chernozhukov et al. (2018)
- Combines the strengths of **machine learning** and **econometrics**



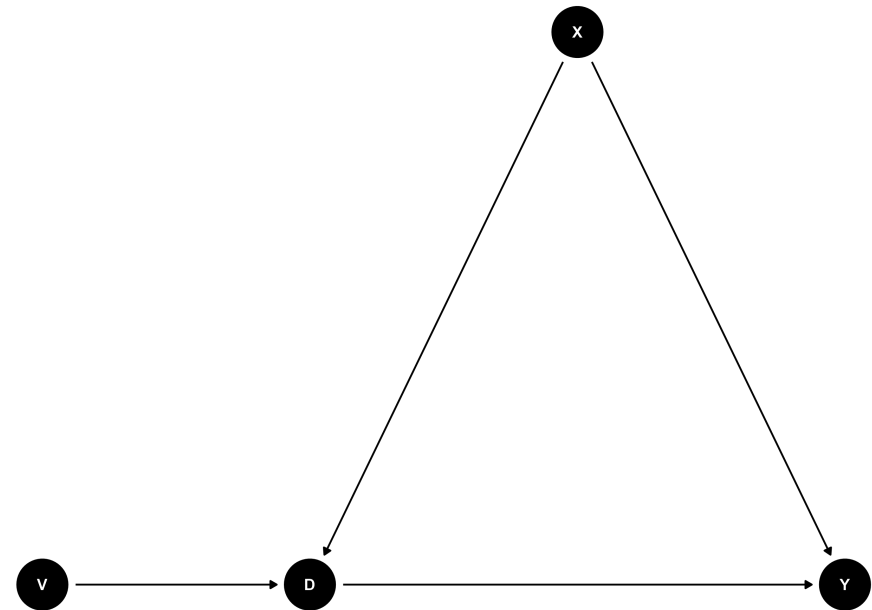
# Motivating Example

## Partially linear regression model (PLR)

$$\begin{aligned} Y &= D\theta_0 + g_0(X) + \zeta, & \mathbb{E}[\zeta|D, X] &= 0, \\ D &= m_0(X) + V, & \mathbb{E}[V|X] &= 0, \end{aligned}$$

with

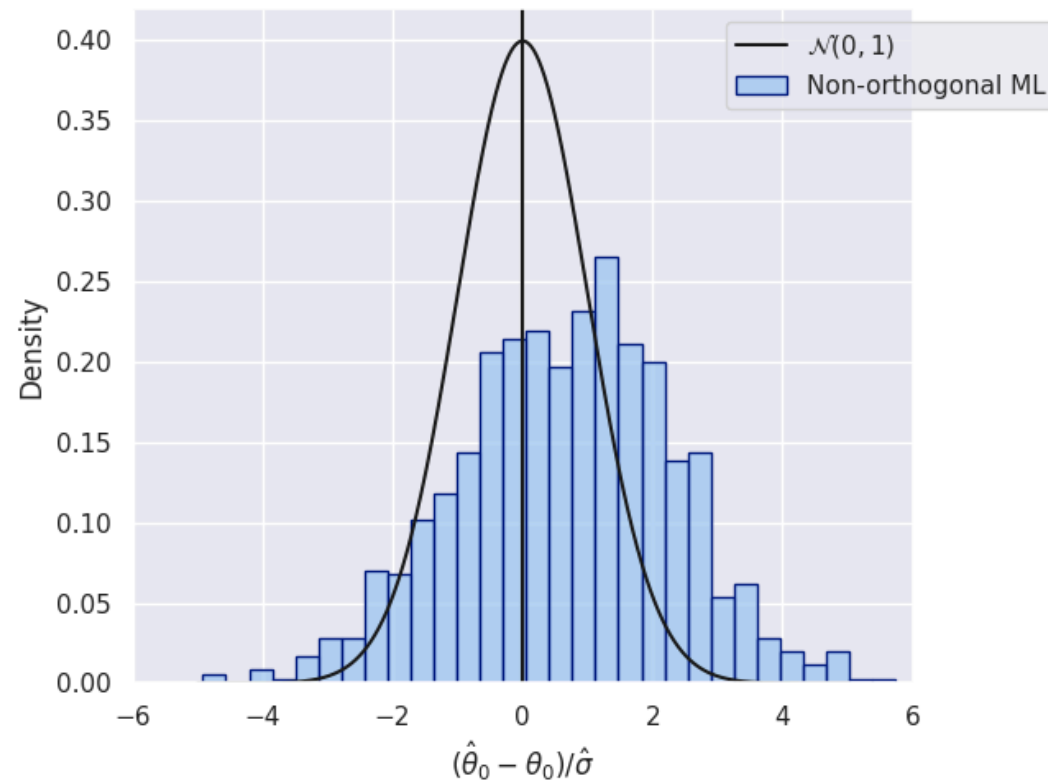
- Outcome variable  $Y$
- Policy or treatment variable of interest  $D$
- High-dimensional vector of confounding covariates  $X = (X_1, \dots, X_p)$
- Stochastic errors  $\zeta$  and  $V$



# Motivating Example

## Failure of naive approach

- What if we simply plug-in ML predictions  $\hat{g}_0(X)$  for  $g_0(X)$  into  $Y = D\theta_0 + g_0(X) + \zeta$ ?
- See this example based on Chernozhukov et al. (2018)



# Motivating Example

## Solution to regularization bias: Orthogonalization

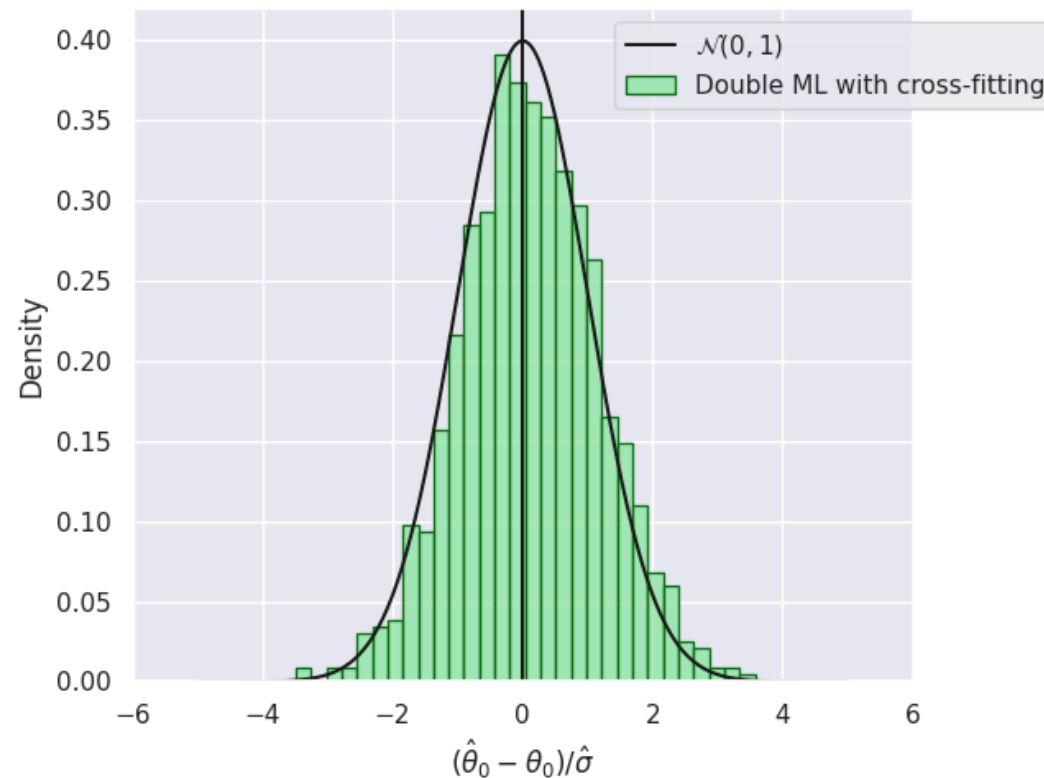
- Remember the *Frisch-Waugh-Lovell* (FWL) Theorem in a **linear regression model** in the low dimensional setting:

$$Y = D\theta_0 + X'\beta + \varepsilon$$

- $\theta_0$  can be consistently estimated by **partialling out**  $X$ , i.e.,
  - OLS regression of  $Y$  on  $X$ :  $\tilde{\beta} = (X'X)^{-1}X'Y \rightarrow \text{Residuals } \hat{\varepsilon}$
  - OLS regression of  $D$  on  $X$ :  $\tilde{\gamma} = (X'X)^{-1}X'D \rightarrow \text{Residuals } \hat{\zeta}$
  - Final OLS regression of  $\hat{\varepsilon}$  on  $\hat{\zeta}$
- Orthogonalization: The idea of the FWL Theorem can be generalized to using ML estimators instead of OLS

# Motivating Example

- Using an orthogonal score leads to an asymptotically normal estimator  $\hat{\theta}_0$
- See this example based on Chernozhukov et al. (2018)





# Neyman Orthogonality

The naive approach minimizes the following MSE

$$\min_{\theta} \mathbb{E}[(Y - D\theta_0 - g_0(X))^2]$$

This implies the following moment equation

$$\mathbb{E}[\underbrace{(Y - D\theta_0 - g_0(X))}_{=: \psi(W, \theta_0, \eta)} \underbrace{D}_{=: \psi(W, \theta_0, \eta)}] = 0$$

Whereas for the partialling-out approach minimizes

$$\min_{\theta} \mathbb{E} \left[ \left( Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta \right)^2 \right]$$

which implies

$$\mathbb{E} \left[ \underbrace{\left( Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta \right)}_{=: \psi(W, \theta_0, \eta)} \underbrace{(D - \mathbb{E}[D|X])}_{=: \psi(W, \theta_0, \eta)} \right] = 0$$

# Neyman Orthogonality

## Naive approach

$$\psi(W, \theta_0, \eta) = (Y - D\theta_0 - g_0(X))D$$

Regression adjustment score

$$\begin{aligned}\eta &= g(X), \\ \eta_0 &= g_0(X).\end{aligned}$$

## FWL partialling out

$$\begin{aligned}\psi(W, \theta_0, \eta_0) &= \left( (Y - E[Y|X]) \right. \\ &\quad \left. - (D - E[D|X])\theta_0 \right) \\ &\quad (D - E[D|X])\end{aligned}$$

Neyman-orthogonal score (Frisch-Waugh-Lovell)

$$\begin{aligned}\eta &= (\ell(X), m(X)), \\ \eta_0 &= (\ell_0(X), m_0(X)), \\ &= (\mathbb{E}[Y | X], \mathbb{E}[D | X]).\end{aligned}$$

# Introduction to Double Machine Learning

# DML Key Ingredients

## 1. Neyman Orthogonality

- Inference is based on a moment condition that satisfies the **Neyman orthogonality condition**  $\psi(W; \theta, \eta)$

$$E[\psi(W; \theta_0, \eta_0)] = 0,$$

where  $W := (Y, D, X, Z)$  and with  $\theta_0$  being the unique solution that obeys the **Neyman orthogonality condition**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0$$

- $\partial_\eta$  denotes the pathwise (Gateaux) derivative operator

# DML Key Ingredients

## 1. Neyman Orthogonality

- **Neyman orthogonality** ensures that the **moment condition** identifying  $\theta_0$  is **insensitive to small perturbations** of the nuisance function  $\eta$  around  $\eta_0$
- Using a Neyman-orthogonal score **eliminates the first order biases** arising from the replacement of  $\eta_0$  with a ML estimator  $\hat{\eta}_0$
- PLR example: Partialling-out score function

$$\psi(\cdot) = (Y - E[Y|X] - \theta(D - E[D|X]))(D - E[D|X])$$

- IRM example: Doubly robust score

$$\begin{aligned} \psi(\cdot) = & g(1, X) - g(0, X) \\ & + \frac{D(Y - g(1, X))}{m(X)} - \frac{(1 - D)(Y - g(0, X))}{1 - m(x)} \\ & - \theta, \end{aligned}$$

# DML Key Ingredients

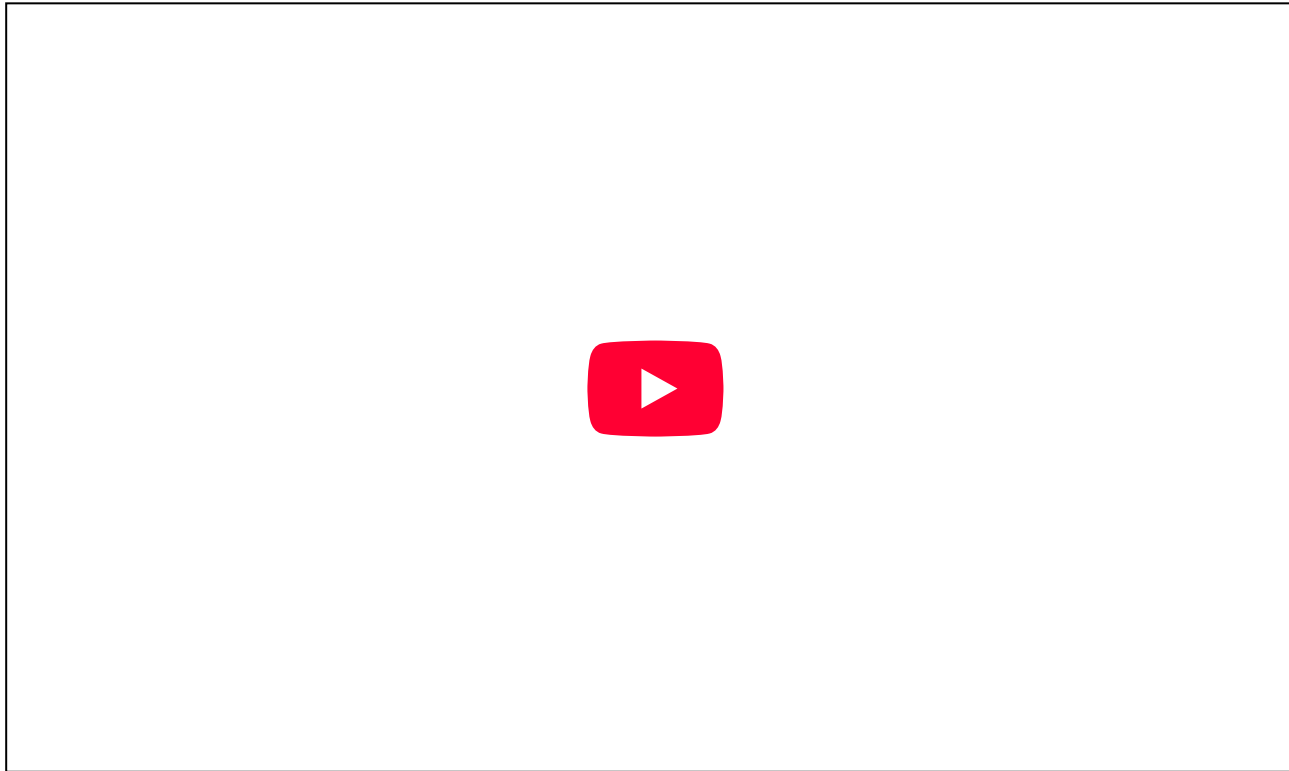
## 2. High-Quality Machine Learning Estimators

- The nuisance parameters are estimated with high-quality (fast-enough converging) machine learning methods
- Different structural assumptions on  $\eta_0$  lead to the use of different machine-learning tools for estimating  $\eta_0$  Chernozhukov et al. (2018) (Section 3)
- Rate requirements depend on the causal model and orthogonal score, e.g. (see Chernozhukov et al. (2018)),
  - PLR, partialling out:  $\|\hat{m}_0 - m_0\|_{P,2} \times (\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}) \leq \delta_N N^{-1/2}$
  - IRM/DR score, ATE:  $\|\hat{m}_0 - m_0\|_{P,2} \times \|\hat{\ell}_0 - \ell_0\|_{P,2} \leq \delta_N N^{-1/2}$

# DML Key Ingredients

## 3. Sample Splitting

- To avoid the biases arising from overfitting, a form of **sample splitting** is used at the stage of producing the estimator of the main parameter  $\theta_0$
- Efficiency gains by using cross fitting (swapping roles of samples for train / hold-out)



# DML Key Ingredients

## Main result in Chernozhukov et al. (2018)

There exist regularity conditions, such that the DML estimator  $\tilde{\theta}_0$  concentrates in a  $1/\sqrt{N}$ -neighborhood of  $\theta_0$  and the sampling error is approximately

$$\sqrt{N}(\tilde{\theta}_0 - \theta_0) \sim N(0, \sigma^2),$$

with

$$\begin{aligned}\sigma^2 &:= J_0^{-2} \mathbb{E}(\psi^2(W; \theta_0, \eta_0)), \\ J_0 &= \mathbb{E}(\psi_a(W; \eta_0)).\end{aligned}$$

- See this example based on Chernozhukov et al. (2018)



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# References

# References

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